Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *DRBENCHMARK* | ***Context:*** *French Pharmaceutical Regulatory Affairs — Document Compliance NLP*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotator profiles are documented as clinical professionals, NLP researchers, and one medical expert (DiaMed) [Q45]. CAS POS is silver-standard via Tagex 3 [Q42]; ESSAI POS is silver-standard via TreeTagger [Q43]. CLISTER STS averages multiple annotator scores [Q41]; DiaMed reports IAA via Cohen's Kappa and Gwet's AC1 [Q46]. Crucially, no annotator with regulatory affairs, pharmacovigilance, or EMA/ANSM submission expertise is identified in any DrBenchmark dataset (cross-cutting CRITICAL [OC] dataset finding), creating a convergent validity violation: the labels reflect clinical-research norms that may systematically diverge from regulatory-legal compliance standards. Dataset analysis surfaces specific cases where this divergence is observable: CLISTER scores 4.0 on quantitatively different lab values (CLISTER-D17, CLISTER-D18) — a clinical annotator's intuition not a regulatory-equivalence judgment; DEFT2020 annotator divergence on precautionary qualifier removal (scores [5.0, 2.0, 4.0, 4.0, 5.0] in DEFT2020-D18, DEFT2020-D23) suggests at least one annotator may have been reading regulatorily but most were not; FrenchMedMCQA pharmacy exam keys may lag current ANSM/EMA standards (FRENCHMEDMCQA-D20). Web search confirmed no published crosswalk between DrBenchmark schemas and regulatory annotation guidelines [WEB-12], and no regulatory affairs annotators identified [WEB-14]. DiaMed IAA values not retrievable from search [WEB-2 access required].

ID	Evidence
Q45	“DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session.” (p.4)
Q42	“CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively.” (p.4)
Q43	“ESSAI (Dalloux et al., 2021) contains 7,247 clinical trial protocols annotated in 41 POS tags using TreeTagger (Schmid, 1994). As the dataset was not originally divided into 3 subsets, we applied the same procedure as on the CAS corpus.” (p.4)
Q41	“CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided.” (p.4)
Q46	“Table 4: Inter-annotator agreement statistics. κ is referring to Kappa Cohen and G to Gwet's AC1.” (p.4)
WEB-12	No French regulatory NER benchmark exists; English ADE Eval required modification before regulatory use (https://pubmed.ncbi.nlm.nih.gov/29860093/)
WEB-14	No regulatory affairs expert annotators identified in any DrBenchmark dataset annotation team (https://arxiv.org/html/2402.13432v1)
WEB-2	DiaMed IAA Table 4 numerical values available in full paper PDF but not surfaced in indexed text (https://aclanthology.org/2024.lrec-main.478/)
CLISTER-D17	Le recul est de 2 ans." / "Le suivie est de 4 ans et demi. — Different follow-up durations scored 4.0; same rubric would conflate regulatory dose differences
CLISTER-D18	L'AHG urinaire est à 10 mmol/l." / "L'AHG urinaire est à 20 mmol/l. — Twofold quantitative difference scored 4.0; regulatory dose doubling requires near-0

DEFT2020-D18	scores=[5.0, 2.0, 4.0, 4.0, 5.0] — Precautionary qualifier dropped in cible; rated 4.0 — legally significant difference not captured
DEFT2020-D23	scores=[5.0, 2.0, 4.0, 4.0, 5.0] — Annotator scoring 2.0 may reflect regulatory-aware reading of precautionary qualifier; others score 4–5
FRENCHMED MCQA-D20	La vaccination contre l'hépatite B: Est obligatoire pour tout le personnel de santé / Est recommandée pour tous les sujets à risque... — Regulatory obligation status; exam ground truth may lag ANSM updates

Strengths

- DEFT2021 NER and MLC are manually annotated (gold standard) [Q25]
- DiaMed IAA documented across two annotation sessions with Cohen's Kappa and Gwet's AC1 [Q45, Q46]
- CLISTER averages multiple annotator scores per pair [Q41]
- DEFT2020 retains individual annotator scores enabling uncertainty quantification (DEFT2020-D12, DEFT2020-D13) — supports the deployment's confidence-score design

ID	Evidence
Q25	“This dataset is manually annotated in two tasks: (i) multi-label classification and (ii) NER.” (p.3)
Q45	“DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session.” (p.4)
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DEFT2020-D12	scores=[5.0, 2.0, 0.0, 3.0, 1.0] — High inter-annotator disagreement; signals annotation uncertainty
DEFT2020-D13	scores=[2.0, 4.5, 0.0, 1.0, 0.0] — Extreme annotator divergence on low-similarity pair

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — ground truth labels reflect clinical/research perspectives, not regulatory affairs perspectives. Confirmed across all 11 datasets via dataset analysis cross-cutting CRITICAL [OC] finding and web search [WEB-14].

OC-2: Likely systematic disagreement on borderline cases. Dataset analysis evidence: DEFT2020-D18 shows one annotator scoring 2.0 vs. others scoring 4–5 on precautionary qualifier removal — consistent with a regulatory-aware annotator weighting differently. CLISTER-D19 / CLISTER-D20 show clinical-intuition scoring of drug dosage comparisons that regulatory annotators would weight differently.

OC-3: Documentation is partial: paper identifies clinical professionals, NLP researchers, and one medical expert for DiaMed [Q45] but does not provide demographic breakdown by region or professional sub-specialty. No regulatory affairs annotators documented anywhere [WEB-14].

OC-4: Re-annotation by regulatory affairs specialists is essentially required for valid deployment evaluation. No such re-annotated subset currently exists [WEB-12].

OC-5: CLISTER averages annotator scores [Q41] which can erase regulatory-aware minority perspectives; DEFT2020 retains individual scores (DEFT2020-D12) which is preferable for uncertainty analysis. Aggregation methods elsewhere not fully documented.

OC-6: Major convergent and external validity concerns: clinical annotation norms produce labels that may systematically diverge from regulatory-legal ground truth. Silver-standard POS in CAS (CAS-D14) and ESSAI (ESSAI-D12) compounds quality concerns. ICD-10 chapter coding in DiaMed reflects hospital reimbursement/epidemiology coding (DIAMED-D4), not regulatory safety axes.

ID	Evidence
WEB-14	No regulatory affairs expert annotators identified in any DrBenchmark dataset annotation team (https://arxiv.org/html/2402.13432v1)
Q45	“DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session.” (p.4)
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DEFT2020-D23	scores=[5.0, 2.0, 4.0, 4.0, 5.0] — Annotator scoring 2.0 may reflect regulatory-aware reading of precautionary qualifier; others score 4–5
CLISTER-D19	Cette patiente, suivie en psychiatrie pour des troubles de l'humeur bipolaires, avait arrêté son traitement au lithium (Teralithe LP® 400 mg, deux comprimés par jour) un mois avant cette ingestion massive. — Named drug with dose in clinical case; regulatory annotator might weight differently
CLISTER-D20	Hépatique (P) T1/2 = 96 h (P) –" / "Hépatique, rénal T1/2 = 15 – 40 jours — Pharmacokinetic comparison; score reflects clinical rather than regulatory judgment
WEB-12	No French regulatory NER benchmark exists; English ADE Eval required modification before regulatory use (https://pubmed.ncbi.nlm.nih.gov/29860093/)
Q41	“CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided.” (p.4)
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CAS-D14	tokens: ['le', 'scanner', 'réalisé', ...] pos_tags: [12, 24, 25, ...] — Automatic POS assignment without full human adjudication — silver-standard risk
ESSAI-D12	Le MEDI9197 est injecté en intra-tumoral tous les 28j (4 semaines). — Drug identifier MEDI9197 tagged as NAM (tag 7) by automatic TreeTagger — silver standard
DIAMED-D4	Il s'agit d'un enfant de 5ans... ayant consulté aux urgences pour asthénie chronique avec pâleur. L'examen clinique a retrouvé un enfant en bon état général avec des tâches achromiques diffuses — Piebaldism case assigned to endocrine chapter; illustrates coarse ICD-10 labeling

Information Gaps

- Specific Cohen's Kappa and Gwet's AC1 values for DiaMed Table 4 — available in paper PDF [WEB-2] but not retrieved

- Annotator demographic breakdown by region (metropolitan vs. overseas) and professional sub-specialty

Requires Expert Verification

- Magnitude of systematic disagreement between clinical annotators and regulatory affairs specialists on borderline STS pairs
- Whether the medical expert annotator in DiaMed has any regulatory affairs experience

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Remediation

Gap 1: No regulatory affairs, pharmacovigilance, or EMA/ANSM submission expert annotators identified in any of the 11 DrBenchmark datasets; ground truth reflects clinical-research norms that may systematically diverge from regulatory-legal standards.

Recommendation: Recruit a small panel of regulatory affairs specialists to re-annotate a stratified sample of borderline STS pairs (especially DEFT2020 and CLISTER drug-leaflet content) and selected QUAERO/emea NER spans, using regulatory-aligned schemas. Compare against existing labels to quantify divergence and surface convergent validity concerns.